Bladogra 50 mg

Mirabegron 50 mg

DOSAGE FORM: Extended release film coated tablets.

COMPOSITION: ***Bladogra 50 mg prolonged-release tablets:*** Each tablet contains 50 mg of mirabegron.

Inactive Ingredients: Polyethylene glycol 20000, polyethylene oxide , hydroxypropylcellulose EF, Butylated hydroxytoluene and Magnesium stearate.

Coat: Hydroxy propyl methyl cellulose E5, Polyethylene glycol 6000, Titanium dioxide, Iron oxide yellow, Iron oxide red and Talc Purified.

PHARMACEUTICAL FORM: Extended release film coated tablet.

CLINICAL PARTICULARS: ***Therapeutic indications:*** Bladogra is indicated for the symptomatic treatment of urgency, increased micturition frequency and/or urgency incontinence as may occur in adult patients with overactive bladder (OAB) syndrome.

POSOLOGY AND METHOD OF ADMINISTRATION: ***Posology Adults (Including Elderly Patients):*** The recommended dose is 50 mg once daily.

SPECIAL POPULATIONS: ***Renal and hepatic impairment:*** Mirabegron has not been studied in patients with end stage renal disease (GFR <15 ml/min/1.73 m^2 or patients requiring haemodialysis) or severe hepatic impairment (Child-Pugh Class C) and it is therefore not recommended for use in these patient populations.

The following table provides the daily dosing recommendations for subjects with renal or hepatic impairment in the absence and presence of strong CYP3A inhibitors.

		Strong CYP3A inhibitors (3)	
		Without inhibitor	With inhibitor
Renal impairment (1)	Mild	50 mg	25 mg
	Moderate	50 mg	25 mg
	Severe	25 mg	Not recommended
Hepatic impairment (2)	Mild	50 mg	25 mg
	Moderate	25 mg	Not recommended

1- Mild: GFR 60 to 89 ml/min/1.73m^2; moderate: GFR 30 to 59 ml/min/1.73 m^2; severe: GFR 15 to 29 ml/min/1.73 m^2. 2- Mild: Child-Pugh Class A; Moderate: Child-Pugh Class B.

Gender: No dose adjustment is necessary according to gender.

Paediatric Population: The safety and efficacy of mirabegron in children below 18 years of age have not yet been established. No data are available.

Method Of Administration: The tablet is to be taken once daily, with liquids, swallowed whole and is not to be chewed, divided, or crushed. It may be taken with or without food.

Contraindications: Mirabegron is contraindicated in patients with: Hypersensitivity to the active substance or to any of the excipients listed.
Severe uncontrolled hypertension defined as systolic blood pressure > 180 mm Hg and/or diastolic blood pressure > 110 mm Hg.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE: ***Renal Impairment:*** Mirabegron has not been studied in patients with end stage renal disease (GFR < 15 ml/min/1.73 m^2) or patients requiring haemodialysis) and, therefore, it is not recommended for use in this patient population. Data are limited in patients with severe renal impairment (GFR 15 to 29 ml/min/1.73 m^2); based on a pharmacokinetic study a dose reduction to 25 mg is recommended in this population. Bladogra is not recommended for use in patients with severe renal impairment (GFR 15 to 29 ml/min/1.73 m^2) concomitantly receiving strong CYP3A inhibitors.

Hepatic Impairment: Mirabegron has not been studied in patients with severe hepatic impairment (Child-Pugh Class C) and, therefore, it is not recommended for use in this patient population. Bladogra is not recommended for use in patients with moderate hepatic impairment (Child-Pugh B) concomitantly receiving strong CYP3A inhibitors.

Hypertension: Mirabegron can increase blood pressure. Blood pressure should be measured at baseline and periodically during treatment with Bladogra, especially in hypertensive patients. Data are limited in patients with stage 2 hypertension (systolic blood pressure ≥ 160 mm Hg or diastolic blood pressure ≥ 100 mm Hg).

Patients With Congenital Or Acquired QT Prolongation: Mirabegron, at therapeutic doses, has not demonstrated clinically relevant QT prolongation in clinical studies. However, since patients with a known history of QT prolongation or patients who are taking medicinal products known to prolong the QT interval were not included in these studies, the effects of mirabegron in these patients is unknown. Caution should be exercised when administering mirabegron in these patients.

Patients With Bladder Outlet Obstruction And Patients Taking Antimuscarinics Medicinal Products For OAB: Urinary retention in patients with bladder outlet obstruction (BOO) and in patients taking antimuscarinic medications for the treatment of OAB has been reported in postmarketing experience in patients taking mirabegron. A controlled clinical safety study in patients with BOO did not demonstrate increased urinary retention in patients treated with mirabegron; however, mirabegron should be administered with caution to patients with clinically significant BOO. Mirabegron should also be administered with caution to patients taking antimuscarinic medications for the treatment of OAB.

Interaction With Other Medicinal Products And Other Forms Of Interaction:

In Vitro data: Mirabegron is transported and metabolized through multiple pathways. Mirabegron is a substrate for cytochrome P450 (CYP) 3A4, CYP2D6, butyrylcholinesterase, uridine diphospho-glucuronosyltransferases (UGT), the efflux transporter P-glycoprotein (P-gp) and the influx organic cation transporters (OCT) OCT1, OCT2, and OCT3. Studies of mirabegron using human liver microsomes and recombinant human CYP enzymes showed that mirabegron is a moderate and time-dependent inhibitor of CYP2D6 and a weak inhibitor of CYP3A. Mirabegron inhibited P-gp-mediated drug transport at high concentrations.

In Vivo data:

DRUG-DRUG INTERACTIONS: The effect of co-administered medicinal products on the pharmacokinetics of mirabegron and the effect of mirabegron on the pharmacokinetics of other medicinal products was studied in single and multiple dose studies. Most drug-drug interactions were studied using a dose of 100 mg mirabegron given as oral controlled absorption system (OCAS) tablets. Interaction studies of mirabegron with metoprolol and with metformin used mirabegron immediate-release (IR) 160 mg. Clinically relevant drug interactions between mirabegron and medicinal products that inhibit, induce or are a substrate for one of the CYP isozymes or transporters are not expected except for the inhibitory effect of mirabegron on the metabolism of CYP2D6 substrates.

Effect Of Enzyme Inhibitors: Mirabegron exposure (AUC) was increased 1.8-fold in the presence of the strong inhibitor of CYP3A/P-gp ketoconazole in healthy volunteers. No dose-adjustment is needed when mirabegron is combined with inhibitors of CYP3A and/or P-gp. However, in patients with mild to moderate renal impairment (GFR 30 to 89 ml/min/1.73 m^2) or mild hepatic impairment (Child- Pugh Class A) concomitantly receiving strong CYP3A inhibitors, such as itraconazole, ketoconazole, ritonavir and clarithromycin, the recommended dose is 25 mg once daily with or without food. Bladogra is not recommended in patients with severe renal impairment (GFR 15 to 29 ml/min/1.73 m^2) or patients with moderate hepatic impairment (Child-Pugh Class B) concomitantly receiving strong CYP3A inhibitors.

Effect Of Enzyme Inducers: Substances that are inducers of CYP3A or P-gp decrease the plasma concentrations of mirabegron. No dose adjustment is needed for mirabegron when administered with therapeutic doses of rifampicin or other CYP3A or P-gp inducers.

Effect of CYP2D6 Polymorphism: CYP2D6 genetic polymorphism has minimal impact on the mean plasma exposure to to mirabegron (see section 5.2). Interaction of mirabegron with a known CYP2D6 inhibitor is not expected and was not studied. No dose adjustment is needed for mirabegron when administered with CYP2D6 inhibitors or in patients who are CYP2D6 poor metabolizers.

Effect Of Mirabegron On CYP2D6 Substrates: In healthy volunteers, the inhibitory potency of mirabegron towards CYP2D6 is moderate and the CYP2D6 activity recovers within 15 days after discontinuation of mirabegron. Multiple once daily dosing of mirabegron IR resulted in a 90% increase in C_{max} and a 229% increase in AUC of a single dose of metoprolol. Multiple once daily dosing of mirabegron resulted in a 79% increase in C_{max} and a 241% increase in AUC of a single dose of desipramine.
Caution is advised if mirabegron is co-administered with medicinal products with a narrow therapeutic index and significantly metabolized by CYP2D6, such as thioridazine, Type 1C antiarrhythmics (e.g., flecainide, propafenone) and tricyclic antidepressants (e.g., imipramine, desipramine). Caution is also advised if mirabegron is co-administered with CYP2D6 substrates that are individually dose titrated.

Effect Of Mirabegron On Transporters: Mirabegron is a weak inhibitor of P-gp. Mirabegron increased C_{max} and AUC by 29% and 27%, respectively, of the P-gp substrate digoxin in healthy volunteers. For patients who are initiating a combination of Mirabegron and digoxin, the lowest dose for digoxin should be prescribed initially. Serum digoxin concentrations should be monitored and used for titration of the digoxin dose to obtain the desired clinical effect. The potential for inhibition of P-gp by mirabegron should be considered when mirabegron is combined with sensitive P-gp substrates e.g. dabigatran.

Other Interactions: No clinically relevant interactions have been observed when mirabegron was co-administered with therapeutic doses of solifenacin, tamsulosin, warfarin, metformin or a combined oral contraceptive medicinal product containing ethinylestradiol and levonorgestrel. Dose-adjustment is not recommended. Increases in mirabegron exposure due to drug-drug interactions may be associated with increases in pulse rate.

FERTILITY, PREGNANCY AND LACTATION: ***Women Of Childbearing Potential:*** Bladogra is not recommended in women of childbearing potential not using contraception.

Pregnancy: There are limited amount of data from the use of mirabegron in pregnant women. Studies in animals have shown reproductive toxicity. This medicinal product is not recommended during pregnancy.

Breast-feeding: No studies have been conducted to assess the impact of mirabegron on milk production in humans, its presence in human breast milk, or its effects on; the breast-fed child. Bladogra should not be administered during breast-feeding.

Fertility: There were no treatment related effects of mirabegron on fertility in animals. The effect of mirabegron on human fertility has not been established.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES: Bladogra has no or negligible influence on the ability to drive and use machines.

UNDESIRABLE EFFECTS: ***Summary Of The Safety Profile:*** The safety of mirabegron was evaluated in 8433 patients with OAB, of which 5648 received at least one dose of mirabegron in the phase 2/3 clinical program, and 622 patients received mirabegron for at least 1 year (365 days). In the three 12-week phase 3 double blind, placebo controlled studies, 88% of the patients completed treatment with mirabegron, and 4% of the patients discontinued due to adverse events. Most adverse reactions were mild to moderate in severity. The most common adverse reactions reported for patients treated with mirabegron 50 mg during the three 12-week phase 3 double blind, placebo controlled studies are tachycardia and urinary tract infections. The frequency of tachycardia was 1.2% in patients receiving mirabegron 50 mg. Tachycardia led to discontinuation in 0.1% patients receiving mirabegron 50 mg. The frequency of urinary tract infections was 2.9% in patients receiving mirabegron 50 mg. Urinary tract infections led to discontinuation in none of the patients receiving mirabegron 50 mg. Serious adverse reactions included atrial fibrillation (0.2%). Adverse reactions observed during the 1 year (long term) active controlled (muscarinic antagonist) study were similar in type and severity to those observed in the three 12-week phase 3 double blind, placebo controlled studies.

ADVERSE REACTIONS: The adverse reactions observed with mirabegron in the three 12-week phase 3 double blind, placebo controlled studies.
The frequency of adverse reactions is defined as follows: very common (≥ 1/10); common (≥ 1/100 to <1/10); uncommon (≥ 1/1,000 to <1/100). rare (≥ 1/10,000 To <1/1000); very rare (<1/10,000) and not known (cannot be established from the available data).

MedDRA System organ class	***Common***	***Uncommon***	***Rare***	***Very rare***	***Not known***
Infections and infestations	Urinary tract infection	Vaginal infection Cystitis			
Psychiatric disorders					Insomnia*
Nervous system disorders	Headache*, Dizziness*				
Eye disorders			Eyelid oedema		
Cardiac disorders	Tachycardia	Palpitation Atrial fibrillation			
Vascular disorders				Hypertensive crisis*	
Gastrointestinal disorders	Nausea*, Constipation*, Diarrhea*	Dyspepsia Gastritis	Lip oedema		
Skin and subcutaneous tissue disorders		Urticaria, Rash, Rash macular, Rash papular, Pruritus	Leukocytoclastic, Vasculitis, Purpura, Angioedema*		
Musculoskeletal and connective tissue disorders		Joint swelling			
Renal and urinary disorders			Urinary retention *		
Reproductive system and breast disorders		Vulvovaginal pruritus			
Investigations		Blood pressure increased, GGT increased, AST increased, ALT increased			

* observed during post-marketing experience.

OVERDOSE: Mirabegron has been administered to healthy volunteers at single doses up to 400 mg. At this dose, adverse events reported included palpitations (1 of 6 subjects) and increased pulse rate exceeding 100 beats per minute (bpm) (3 of 6 subjects). Multiple doses of mirabegron up to 300 mg daily for 10 days showed increases in pulse rate and systolic blood pressure when administered to healthy volunteers. Treatment for overdose should be symptomatic and supportive. In the event of overdose, pulse rate, blood pressure, and ECG monitoring is recommended.

PHARMACOLOGICAL PROPERTIES: ***Pharmacodynamic Properties:*** Pharmacotherapeutic group: Urologicals, Urinary antispasmodics.

Mechanism Of Action: Mirabegron is a potent and selective beta 3-adrenoceptor agonist. Mirabegron showed relaxation of bladder smooth muscle in rat and human isolated tissue, increased cyclic adenosine monophosphate (cAMP) concentrations in rat bladder tissue and showed a bladder relaxant effect in rat urinary bladder function models. Mirabegron increased mean voided volume per micturition and decreased the frequency of non-voiding contractions, without affecting voiding pressure, or residual urine in rat models of bladder overactivity. In a monkey model, mirabegron showed decreased voiding frequency. These results indicate that mirabegron enhances urine storage function by stimulating beta 3-adrenoceptors in the bladder. During the urine storage phase, when urine accumulates in the bladder, sympathetic nerve stimulation predominates. Noradrenaline is released from nerve terminals, leading predominantly to beta adrenoceptor activation in the bladder musculature, and hence bladder smooth muscle relaxation. During the urine voiding phase, the bladder is predominantly under parasympathetic nervous system control. Acetylcholine, released from pelvic nerve terminals, stimulates cholinergic M2 and M3 receptors, inducing bladder contraction. The activation of the M2 pathway also inhibits beta 3-adrenoceptor induced increases in cAMP. Therefore beta 3- adrenoceptor stimulation should not interfere with the voiding process. This was confirmed in rats with partial urethral obstruction, where mirabegron decreased the frequency of non-voiding contractions without affecting the voided volume per micturition, voiding pressure, or residual urine volume.

Pharmacodynamic Effects: Urodynamics: Mirabegron at doses of 50 mg and 100 mg once daily for 12 weeks in men with lower urinary tract symptoms (LUTS) and bladder outlet obstruction (BOO) showed no effect on cystometry parameters and was safe and well tolerated. The effects of mirabegron on maximum flow rate and detrusor pressure at maximum flow rate were assessed in this urodynamic study consisting of 200 male patients with LUTS and BOO. Administration of mirabegron at doses of 50 mg and 100 mg once daily for 12 weeks did not adversely affect the maximum flow rate or detrusor pressure at maximum flow rate. In this study in male patients with LUTS/BOO, the adjusted mean (SE) change from baseline to end of treatment in post void residual volume (ml) was 0.55 (10.702), 17.89 (10.190), 30.77 (10.598) for the placebo, mirabegron 50 mg and mirabegron 100 mg treatment groups.

Effect On QT Interval: Mirabegron at doses of 50 mg or 100 mg had no effect on the QT interval individually corrected for heart rate (QTcI interval) when evaluated either by sex or by the overall group.

Effects On Pulse Rate And Blood Pressure In Patients With OAB: In OAB patients (mean age of 59 years) across three 12-week phase 3 double blind, placebo controlled studies receiving mirabegron 50 mg once daily an increase in mean difference from placebo of approximately 1 bpm for pulse rate and approximately 1 mm Hg or less in systolic blood pressure/ diastolic blood pressure (SBP/DBP) was observed. Changes in pulse rate and blood pressure are reversible upon discontinuation of treatment.

Effect on intraocular pressure (IOP): Mirabegron 100 mg once daily did not increase IOP in healthy subjects after 56 days of treatment. In a phase 1 study assessing the effect of mirabegron on IOP using Goldmann applanation tonometry in 310 healthy subjects, a dose of mirabegron 100 mg was non-inferior to placebo for the primary endpoint of the treatment difference in mean change from baseline to day 56 in subject-average IOP; the upper bound of the two-sided 95% CI of the treatment difference between mirabegron 100 mg and placebo was 0.3 mm Hg.

PHARMACOKINETIC PROPERTIES: ***Absorption:*** After oral administration of mirabegron in healthy volunteers mirabegron is absorbed to reach peak plasma concentrations (C_{max}) between 3 and 4 hours. The absolute bioavailability increased from 29% at a dose of 25 mg to 35% at a dose of 50 mg. Mean C_{max} and AUC increased more than dose proportionally over the dose range. In the overall population of males and females, a 2- fold increase in dose from 50 mg to 100 mg mirabegron increased C_{max} and AUC_{tau} by approximately 2.9- and 2.6-fold, respectively, whereas a 4-fold increase in dose from 50 mg to 200 mg mirabegron increased C_{max} and AUC_{tau} by approximately 8.4- and 6.5-fold. Steady state concentrations are achieved within 7 days of once daily dosing with mirabegron. After once daily administration, plasma exposure of mirabegron at steady state is approximately double that seen after a single dose.

Effect of food on absorption: Co-administration of a 50 mg tablet with a high-fat meal reduced mirabegron Cmax and AUC by 45% and 17%, respectively. A low-fat meal decreased mirabegron Cmax and AUC by 75% and 51%, respectively. In the phase 3 studies, mirabegron was administered with or without food and demonstrated both safety and efficacy. Therefore, mirabegron can be taken with or without food at the recommended dose.

Distribution: Mirabegron is extensively distributed. The volume of distribution at steady state is approximately 1670 L. Mirabegron is bound (approximately 71%) to human plasma proteins, and shows moderate affinity for albumin and alpha-1 acid glycoprotein. Mirabegron distributes to erythrocytes. In vitro erythrocyte concentrations of ^{l4}C-mirabegron were about 2-fold higher than in plasma.

Biotransformation: Mirabegron is metabolized via multiple pathways involving dealkylation, oxidation, (direct) glucuronidation, and amide hydrolysis. Mirabegron is the major circulating component following a single dose of ^{l4}C-mirabegron. Two major metabolites were observed in human plasma; both are phase 2 glucuronides representing 16% and 11% of total exposure. These metabolites are not pharmacologically active.

Elimination: Total body clearance (CL_{tot}) from plasma is approximately 57 L/h. The terminal elimination half-life (t1/2) is approximately 50 hours. Renal clearance (CLR) is approximately 13 L/h, which corresponds to nearly 25% of $CLCL_{tot}$. Renal elimination of mirabegron is primarily through active tubular secretion along with glomerular filtration. The urinary excretion of unchanged mirabegron is dose-dependent and ranges from approximately 6.0% after a daily dose of 25 mg to 12.2% after a daily dose of 100 mg. Following the administration of 160 mg ^{l4}C-mirabegron to healthy volunteers, approximately 55% of the radiolabel was recovered in the urine and 34% in the faeces. Unchanged mirabegron accounted for 45% of the urinary radioactivity, indicating the presence of metabolites. Unchanged mirabegron accounted for the majority of the faecal radioactivity.

Age: The C_{max} and AUC of mirabegron and its metabolites following multiple oral doses in elderly volunteers (> 65 years) were similar to those in younger volunteers (18-45 years).

Gender: The C_{max} and AUC are approximately 40% to 50% higher in females than in males. Gender differences in C_{max} and AUC are attributed to differences in body weight and bioavailability.

Race: The pharmacokinetics of mirabegron are not influenced by race.

Renal impairment: Following single dose administration of 100 mg mirabegron in volunteers with mild renal impairment (eGFR-MDRD 60 to 89 ml/min/1.73 m^2), mean mirabegron C_{max} and AUC were increased by 6% and 31% relative to volunteers with normal renal function. In volunteers with moderate renal impairment (eGFR-MDRD 30 to 59 ml/min/1.73 m^2), C_{max} and AUC were increased by 23% and 66%, respectively. In volunteers with severe renal impairment (eGFR- MDRD 15 to 29 ml/min/1.73 m^2), mean C_{max} and AUC values were 92% and 118% higher. Mirabegron has not been studied in patients with end stage renal disease (GFR < 15 ml/min/1.73 m^2 or patients requiring haemodialysis).

Hepatic impairment: Following single dose administration of 100 mg mirabegron in volunteers with mild hepatic impairment (Child-Pugh Class A), mean mirabegron C_{max} and AUC were increased by 9% and 19% relative to volunteers with normal hepatic function. In volunteers with moderate hepatic impairment (Child-Pugh Class B), mean C_{max} and AUC values were 175% and 65% higher. Mirabegron has not been studied in patients with severe hepatic impairment (Child-Pugh Class C).

SPECIAL PRECAUTIONS FOR STORAGE: Store in a temperature not exceeding 30°C in a dry place.

Nature And Contents Of Container: Carton box containing (Al/Al) strips each of 10 film coated tablets + inner leaflet.

Keep All Medicines Out of Reach of Children

Manufactured by: Apex Pharma - S.A.E - Badr City- Egypt.
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